The documentation and process conversion measures necessary to comply with this revision shall be completed by 6 May 2024.

INCH-POUND

MIL-PRF-19500/592H
w/AMENDMENT 2
6 February 2024
SUPERSEDING
MIL-PRF-19500/592H
w/AMENDMENT 1
11 March 2021

PERFORMANCE SPECIFICATION SHEET

TRANSISTOR, REPETITIVE AVALANCHE, FIELD EFFECT, N-CHANNEL, SILICON, TYPES 2N7224, 2N7225, 2N7227, 2N7228, JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for an N-channel, enhancement-mode, MOSFET, power transistor intended for use in high density power switching applications. Four levels of product assurance (JAN, JANTX, JANTXV, and JANS) are provided for each encapsulated device type. Two levels of product assurance (JANHC and JANKC) are provided for each unencapsulated device type die, with avalanche energy ratings (E_{AS} and E_{AR}) and maximum avalanche current (I_{AR}).

1.2 Package outlines. The device package outlines are as follows: TO-254AA in accordance with figure 1 and TO-276AB in accordance with figure 2 for all encapsulated device types. See figures 3, 4, and 5 for unencapsulated devices.

1.3 Maximum ratings. (T_A = +25°C, unless otherwise specified).

Type	P_T (1) T_C = +25°C	P_T T_A = +25°C	V_{GS}	I_{D1} (2) (3) T_C = +25°C	I_{D2} (2) T_C = +100°C	I_S	I_{DM} (4)	T_J and T_{STG}	V_{ISO} at 70,000 foot	$R_{\theta JC}$ max (5)
	W	W	V dc	A dc	A dc	A dc	A(pk)	°C		°C/W
2N7224, 2N7224U	150	4.0	±20	34.0	21	34.0	136	-55		0.83
2N7225, 2N7225U	150	4.0	±20	27.4	17	27.4	110	to		0.83
2N7227, 2N7227U	150	4.0	±20	14.0	9	14.0	56	+150	400	0.83
2N7228, 2N7228U	150	4.0	±20	12.0	8	12.0	48		500	0.83

See notes on next page.

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1.3 Maximum ratings - Continued.

Type	I_{AR}	E_{AS}	E_{AR}	$r_{DS(on)}$ max (1) (6) V_{GS} = 10 V dc $I_D = I_{D2}$	
				T_J = +25°C	T_J = +150°C
	A	mj	mj	Ω	Ω
2N7224, 2N7224U	34.0	150	15.0	0.070	0.133
2N7225, 2N7225U	27.4	500	15.0	0.100	0.200
2N7227, 2N7227U	14.0	700	15.0	0.315	0.693
2N7228, 2N7228U	12.0	750	15.0	0.415	0.913

(1) Derate linearly 1.2 W/°C for T_C > +25°C.
(2) The following formula derives the maximum theoretical I_D limit. I_D is limited by package and internal wires and may be limited by pin diameter:

$$I_D = \sqrt{\frac{T_{JM} - T_C}{(R_{\theta JC}) \times (R_{DS}(on) \text{ at } T_{JM})}}$$

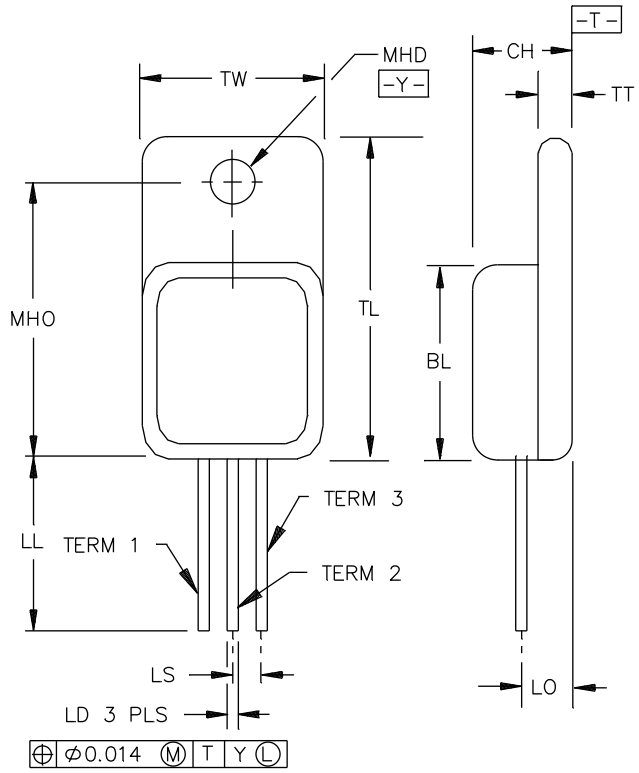
(3) See figure 6, maximum drain current graph.
(4) I_{DM} = 4 X I_{D1} as calculated in note (2).
(5) See figure 7, thermal impedance curves.
(6) Pulsed (see 4.5.1).

1.4 Primary electrical characteristics. T_C = +25°C (unless otherwise specified).

Type	Min $V_{(BR)DSS}$ V_{GS} = 0 I_D = 1.0 mA dc	$V_{GS(th)1}$ $V_{DS} \geq V_{GS}$ I_D = 0.25 mA		Max I_{DSS1} V_{GS} = 0 V_{DS} = 80 percent of rated V_{DS}	Max $r_{DS(on)}$ (1) V_{GS} = 10 V dc $I_D = I_{D2}$ T_J = +25°C
	V dc	V dc		μA dc	Ohms
		Min	Max		
2N7224, 2N7224U	100	2.0	4.0	25	0.070
2N7225, 2N7225U	200	2.0	4.0	25	0.100
2N7227, 2N7227U	400	2.0	4.0	25	0.315
2N7228, 2N7228U	500	2.0	4.0	25	0.415

(1) Pulsed (see 4.5.1).

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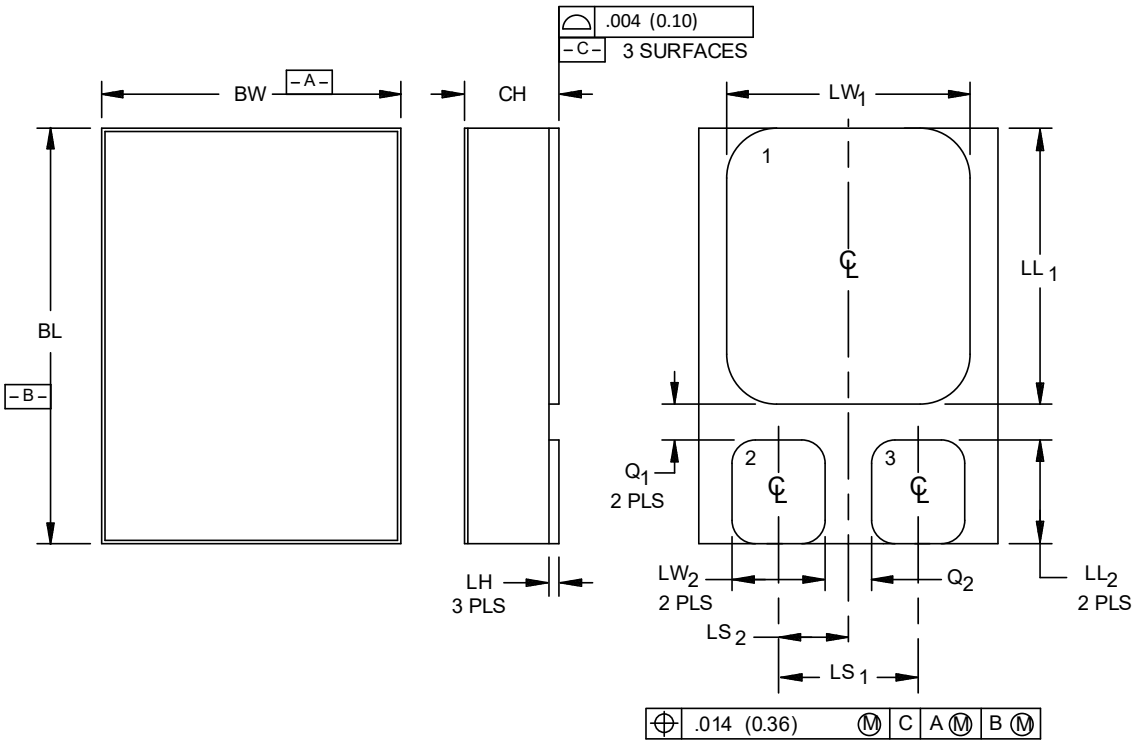


Ltr.	Dimension				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
BL	.535	.545	13.59	13.84	
CH	.249	.260	6.32	6.60	
LD	.035	.045	0.89	1.14	
LL	.510	.570	12.95	14.48	
LO	.150 BSC		3.81 BSC		
LS	.150 BSC		3.81 BSC		
MHD	.139	.149	3.53	3.78	
MHO	.665	.685	16.89	17.40	
TL	.790	.800	20.07	20.32	3, 4
TT	.040	.050	1.02	1.27	
TW	.535	.545	13.59	13.84	3, 4
Term 1	Drain				
Term 2	Source				
Term 3	Gate				

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
* 3. Glass meniscus included in dimension TL and TW.
4. All terminals are isolated from the case.

FIGURE 1. Physical dimensions for TO-254AA.

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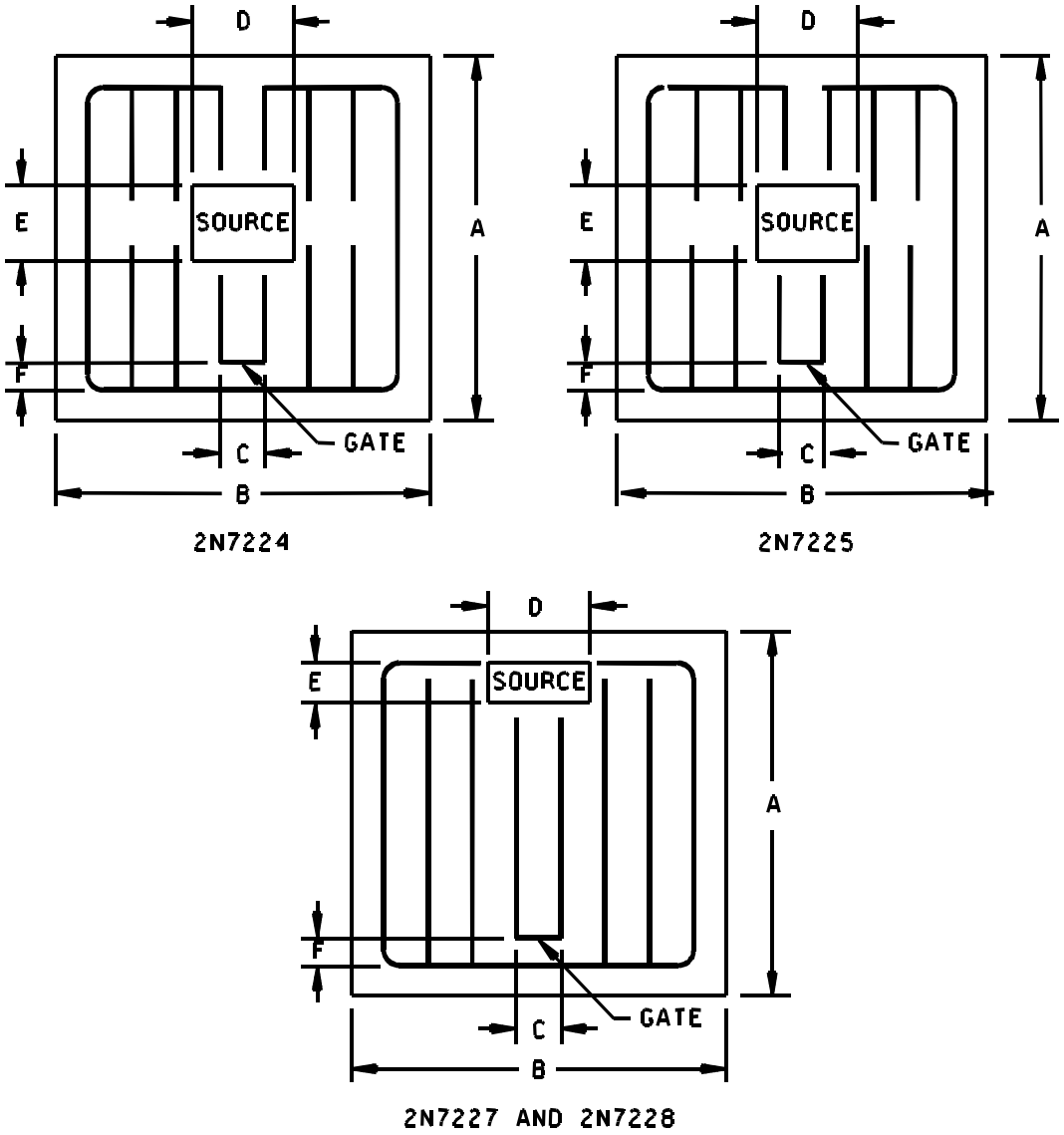


Dimensions				
Ltr.	Inches		Millimeters	
	Min	Max	Min	Max
BL	.620	.630	15.75	16.00
BW	.445	.455	11.30	11.56
CH		.142		3.60
LH	.010	.020	0.26	0.50
LL_1	.410	.420	10.41	10.67
LL_2	.152	.162	3.86	4.11
LS_1	.210 BSC		5.33 BSC	
LS_2	.105 BSC		2.67 BSC	
LW_1	.370	.380	9.40	9.65
LW_2	.135	.145	3.43	3.68
Q_1	.030		0.76	
Q_2	.035		0.89	
Term 1	Drain			
Term 2	Gate			
Term 3	Source			

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for information only.
3. The lid shall be electrically isolated from the drain, gate and source.

FIGURE 2. Dimensions and configuration of surface mount package outline (TO-276AB) 2N7224U,2N7225U, 2N7227U, and 2N7228U.

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NOTES:

1. Dimensions are in inches.
2. Millimeters are given for information only.
3. Unless otherwise specified, tolerance is ±.005 inches (0.13 mm).
4. Physical characteristics of the die thickness = .0187 inch (0.47 mm).
5. Back metal: Cr - Ni - Ag.
6. Top metal: Al.
7. Back contact: Drain.

FIGURE 3. Physical dimensions (A-version) JANHC and JANKC.

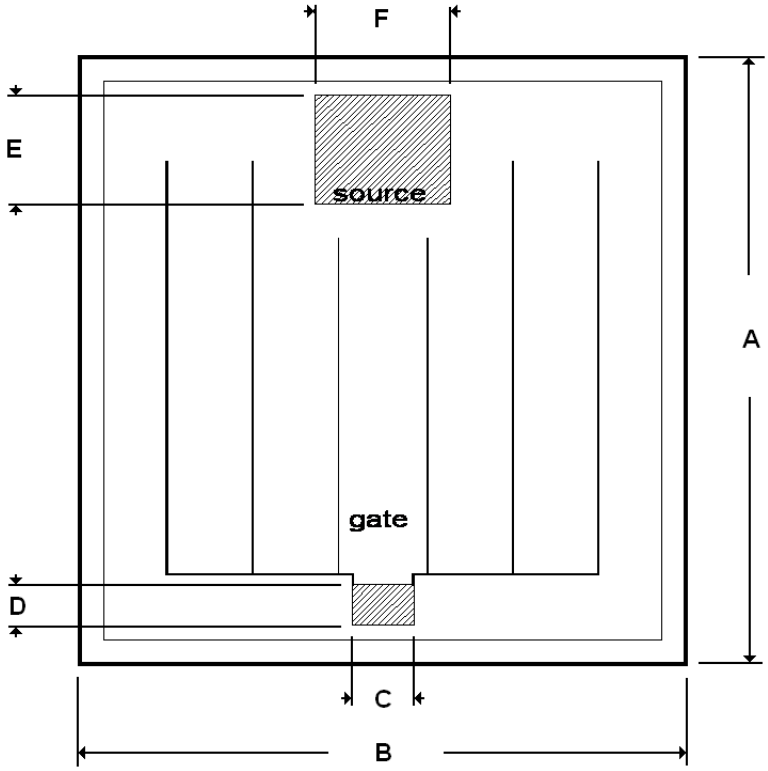
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A version

Ltr	Dimensions - 2N7224 and 2N7225				Dimensions - 2N7227 and 2N7228			
	Inches		Millimeters		Inches		Millimeters	
	Min	Max	Min	Max	Min	Max	Min	Max
A	.2475	.2633	6.287	6.687	.252	.262	6.40	6.65
B	.2479	.2636	6.296	6.696	.252	.262	6.40	6.65
C	.0181	.0184	0.459	0.469	.025	.035	0.64	0.89
D	.0500	.0504	1.270	1.280	.043	.053	1.09	1.35
E	.0315	.0319	0.799	0.809	.032	.042	0.81	1.07
F	.0706	.0710	1.793	1.803	.015	.025	0.38	0.64

FIGURE 3. Physical dimensions (A-version) JANHC and JANKC - Continued.

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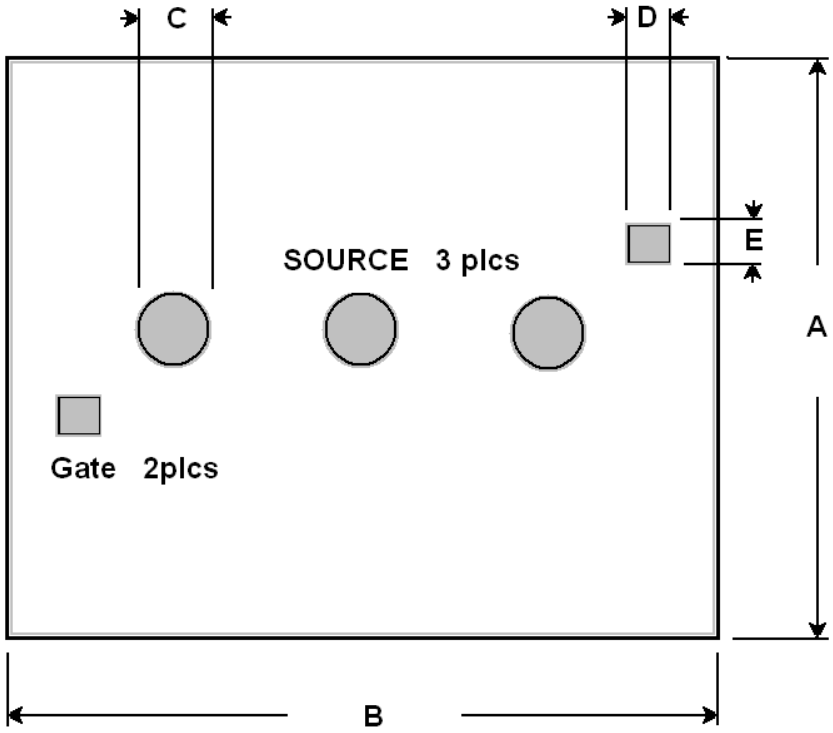
Ltr	Dimensions - 2N7224 and 2N7225			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.254	.260	6.45	6.60
B	.254	.260	6.45	6.60
C	.028	.033	.71	.84
D	.017	.022	.43	.56
E	.047	.053	1.19	1.35
F	.059	.065	1.50	1.65

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 inch (0.13 mm).
4. The physical characteristics of the die are: The back metals are chromium, nickel, and silver and the back contact is the drain. The top metal is aluminum.
5. Die thickness is .015 inch (0.38 mm) ±.001 inch (0.025 mm).

FIGURE 4. JANHC and JANKC (B-version) die dimensions for 2N7224 and 2N7225.

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2N7227 and 2N7228



Ltr	Dimensions - 2N7227 and 2N7228			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.247	.253	6.27	6.43
B	.287	.293	7.29	7.44
C	.033	.037	.84	.94
D	.016	.020	.41	.51
E	.017	.021	.43	.53

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 inch (0.13 mm).
4. The physical characteristics of the die are: The back metals are chromium, nickel, and silver and the back contact is the drain. The top metal is aluminum.
5. Die thickness is .018 inch (0.46 mm) ±.001 inch (0.025 mm).

FIGURE 5. JANHC and JANKC (B-version) die dimensions for 2N7227 and 2N7228.

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1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.5 for PIN construction example and 6.6 for a list of available PINs.

1.5.1 JAN certification mark and quality level for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JAN", "JANTX", "JANTXV" and "JANS".

1.5.2 JAN certification mark and quality level for unencapsulated devices (die). The quality level designators for unencapsulated devices (die) that are applicable for this specification sheet from the lowest to the highest level are as follows: "JANHC" and "JANKC".

1.5.3 Device type. The designation system for the device types of transistors covered by this specification sheet are as follows.

1.5.3.1 First number and first letter symbols. The transistors of this specification sheet use the first number and letter symbols "2N".

1.5.3.2 Second number symbols. The second number symbols for the transistors covered by this specification sheet are as follows: "7224", "7225", “7227”, and "7228".

1.5.3.3 Suffix letters. No suffix letters are used on devices that are packaged in the TO-254AA package of figure 1. The suffix letter "U" is used on devices that are packaged in the TO-276AB package of figure 2.

1.5.4 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.

1.5.5 Die identifiers for unencapsulated devices (manufacturers and critical interface identifiers). The manufacturer die identifiers that are applicable for this specification sheet are "A" and "B" (see figures 3, 4, and 5).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.
* MIL-STD-883 - Test Method Standard for Microcircuits.

(Copies of these documents are available online at https://quicksearch.dla.mil/).

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figures 1 (TO-254AA), 2 (TO-276AB, surface mount), 3, 4, and 5 (die) herein. Methods used for electrical isolation of the terminal feedthroughs shall employ materials that contain a minimum of 90 percent AL_2O_3 (ceramic). Examples of such construction techniques are metallized ceramic eyelets or ceramic walled packages.

3.4.1 Lead formation, material, and finish. Lead material shall be Kovar or Alloy 52; a copper core or plated core is permitted. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead formation material or finish is desired, it shall be specified in the acquisition document (see 6.2). When lead formation is performed, as a minimum, the vendor shall perform 100-percent hermetic seal in accordance with screen 14 of table E-IV of MIL-PRF-19500 and 100-percent dc testing in accordance with table I, subgroup 2 herein.

3.4.2 Internal construction. Multiple chip construction shall not be permitted to meet the requirements of this specification.

* 3.4.3 Silicone die coating. The use of a silicone die coat requires a successful completion of method 5011 of MIL-STD-883 on each lot of silicon die coating for its intended applications.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.6 Electrostatic discharge sensitive (ESDS). The devices covered by this specification sheet have been classified as ESDS. The devices shall be handled in accordance with the ESD program established to comply with the requirements of MIL-PRF-19500 to avoid damage due to the accumulation of static charge. The following handling practices shall be followed:

a. Devices should be handled on benches with conductive and grounded surface.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent if practical.

g. Care should be exercised, during test and troubleshooting, to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source. $R \leq 100\ k\Omega$, whenever bias voltage is to be applied drain to source.

3.7 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.8 Electrical test requirements. The electrical test requirements shall be table I as specified herein.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500. Alternate flow is allowed for qualification inspection in accordance with MIL-PRF-19500.

4.2.1 JANHC and JANKC devices. Qualification shall be in accordance with MIL-PRF-19500.

4.2.2 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

* 4.2.3 Silicone die coating. When silicon die coating is used, method 5011 of MIL-STD-883 shall be performed on that coating in the full MIL-PRF-19500 qualification process.

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4.3 Screening (JANTX, JANTXV, and JANS levels only). Screening shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

*	Screen (1) (2)	Measurements for JANS level	Measurements for JANTX and JANTXV levels
	(3)	Gate stress (see 4.3.2)	Gate stress (see 4.3.2)
	(3)	Method 3470 of MIL-STD-750 (see 4.3.3)	Method 3470 of MIL-STD-750 (see 4.3.3)
	(3) 3c	Method 3161 of MIL-STD-750 (see 4.3.4)	Method 3161 of MIL-STD-750 (see 4.3.4)
*	5	Method 2052 of MIL-STD-750, PIND (see MIL-PRF-19500 and 4.3.6)	Not applicable
	9	I_{GSSF1}, I_{GSSR1}, I_{DSS1}, subgroup 2 of table herein	Subgroup 2 of table I herein.
	10	Method 1042 of MIL-STD-750, test condition B	Method 1042 of MIL-STD-750, test condition B
	11	Subgroup 2 of table I herein I_{GSSF1}, I_{GSSR1}, I_{DSS1}, $r_{DS(on)1}$, $V_{GS(th)1}$; ΔI_{GSSF1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±25 μA dc or ±100 percent of initial value, whichever is greater.	Subgroup 2 of table I herein. I_{GSSF1}, I_{GSSR1}, I_{DSS1}, $r_{DS(on)1}$, $V_{GS(th)1}$
	12	Method 1042 of MIL-STD-750, condition A	Method 1042 of MIL-STD-750, condition A
	13	Subgroups 2 and 3 of table I herein; ΔI_{GSSF1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±25 μA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ±20 percent of initial value, $\Delta V_{GS(th)1}$ = ±20 percent of initial value.	Subgroup 2 of table I herein; ΔI_{GSSF1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±25 μA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ±20 percent of initial value, $\Delta V_{GS(th)1}$ = ±20 percent of initial value.
	17	Method 1081 of MIL-STD-750 (see 4.3.5) Endpoints: Subgroup 2 of table I herein (Not applicable for surface mount devices)	Method 1081 of MIL-STD-750 (see 4.3.5) Endpoints: Subgroup 2 of table I herein (Not applicable for surface mount devices)

(1) At the end of the test program, I_{GSSF1}, I_{GSSR1}, and I_{DSS1} are measured.
(2) An out-of-family program to characterize I_{GSSF1}, I_{GSSR1}, I_{DSS1}, $V_{GS(th)1}$, and $r_{DS(ON)1}$ shall be invoked.
(3) Shall be performed anytime after temperature cycling, screen 3a; and does not need to be repeated in screening requirements.

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4.3.1. Screening (JANHC and JANKC). Screening of JANHC and JANKC die shall be in accordance with MIL-PRF-19500, "Discrete Semiconductor Die/Chip Lot Acceptance". As a minimum, die shall be 100 percent probed in accordance with table I, subgroup 2, except test current shall not exceed 20 A. Burn-in duration for the JANKC level follows JANS requirements; the JANHC follows JANTX requirements.

4.3.2 Gate stress test. Apply V_{GS} = 30 V minimum for t = 250 μs minimum.

4.3.3 Single pulsed unclamped inductive switching.

a. Peak current, I_D.....I_{AR}(max).

b. Peak gate voltage, V_{GS} .....10 V.

c. Gate to source resistor, R_{GS}.....25 ≤ R_g ≤200 Ω.

d. Initial case temperature .....+25°C, +10°C, -5°C.

e. Inductance, L ..... $\left[\frac{2E_{AS}}{(I_{D1})^2}\right]$ $\left[\frac{(V_{BR}-V_{DD})}{V_{BR}}\right]$ mH minimum.

f. Number of pulses to be applied .....1 pulse minimum.

g. Supply voltage (V_{DD}) .....50 V, (25 V for devices with minimum $V_{(BR)DSS}$ of 100 V).

4.3.4 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3161 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{MD}, t_{SW}, (and V_H where appropriate). See table II, group E, subgroup 4 herein and figure 7.

4.3.5 Dielectric withstanding voltage.

a. Magnitude of test voltage……900V DC

b. Duration of application of test voltage……..15 seconds (min)

c. Points of application of test voltage……All leads to case (bunch connection)

d. Method of connection……Mechanical

e. Kilovolt-ampere rating of high voltage source……..1200V/1.0 mA (min)

f. Maximum leakage current……1.0 mA

g. Voltage ramp up time……500V/second

4.3.6 PIND. Not applicable in screening when devices are processed using alternative method and flow requirements approved by the qualifying activity, that includes incorporating the use of certified clean processing and silicone die coat. Instead, the PIND test performance shall be performed in group B3 and group C3, on a lot sample basis. PIND failures detected in group B or C will represent lot jeopardy and shall be evaluated for root cause and lot integrity.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein.

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4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIA (JANS) and table E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500 and as follows.

4.4.2.1 Quality level JANS, table E-VIA of MIL-PRF-19500.

Subgroup	Method	Conditions
B3	1051	Test condition G.
B3	2075	See 3.4.2 herein.
B3	2052	PIND, required if not performed in screening. (22 devices, c = 0 for large lots, 12 devices, c = 0 for small lots).
B3	2037	Test condition A. All internal wires for each device shall be pulled separately. If group B3 is to be continued to C6, strength test may be performed after C6.
B4	1042	Test condition D, 2,000 cycles (6,000 cycles for devices with .008 inch or larger bond wires). The heating cycle shall be 1 minute minimum. No heat sink or forced air cooling on the device shall be permitted during the on-cycle.
B5	1042	A separate sample may be pulled for each test. Accelerated steady-state reverse bias; test condition A, V_{DS} = rated, T_A = +175°C, t = 120 hours, read and record $V_{BR(DSS)}$ (pre and post) at I_D = 1 mA. Read and record I_{DSS} (pre and post) in accordance with table I, subgroup 2 herein. $V_{BR(DSS)}$ delta cannot exceed 10 percent.
B5	1042	Accelerated steady-state gate stress; test condition B, V_{GS} = rated, T_A = +175°C, t = 24 hours.

4.4.2.2 Quality levels JAN, JANTX and JANTXV, table E-VIB of MIL-PRF-19500.

Subgroup	Method	Conditions
B2	1051	Test condition G.
B3	1042	Test condition D, 2,000 cycles. The heating cycle shall be 1 minute minimum. No heat sink or forced air cooling on the device shall be permitted during the on-cycle.
B3	2037	Test condition D. All internal wires for each device shall be pulled separately. If group B3 is to be continued to C6, bond strength test may be performed after C6.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500, and as follows.

Subgroup	Method	Conditions
C2	2036	Test condition A; weight = 10 pounds, t = 15 s (not applicable for surface mount devices).
C3	2052	PIND, required if not performed in screening. (JANS only, 22 devices, c = 0 for large lots, 12 devices, c = 0 for small lots).
C5	3161	See 4.3.4.
C6	1042	Test condition D, 6,000 cycles. The heating cycle shall be 1 minute minimum. No heat sink or forced air cooling on the device shall be permitted during the on-cycle.

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4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-IX of MIL-PRF-19500 and as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Thermal impedance 2/	3161	See 4.3.4.	$Z_{\theta JC}$			°C/W
Breakdown voltage, drain to source	3407	I_D = 1.0 mA dc, bias condition C, V_{GS} = 0 V dc	$V_{(BR)DSS}$			
2N7224, 2N7224U				100		V dc
2N7225, 2N7225U				200		V dc
2N7227, 2N7227U				400		V dc
2N7228, 2N7228U				500		V dc
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$; I_D = .25 mA	$V_{GS(th)1}$	2.0	4.0	V dc
Gate reverse current	3411	V_{GS} = +20 V dc, bias condition C, V_{DS} = 0	I_{GSSF1}		+100	nA dc
Gate reverse current	3411	V_{GS} = -20 V dc, bias condition C, V_{DS} = 0	I_{GSSR1}		-100	nA dc
Drain current	3413	V_{DS} = 80 percent of rated V_{DS}, bias condition C, V_{GS} = 0	I_{DSS1}		25	µA dc
Static drain to source on-state resistance	3421	V_{GS} = 10 V dc, condition A, pulsed (see 4.5.1), I_D = rated I_{D2} (see 1.3)	$r_{DS(on)1}$			
2N7224, 2N7224U					0.070	Ohm
2N7225, 2N7225U					0.100	Ohm
2N7227, 2N7227U					0.315	Ohm
2N7228, 2N7228U					0.415	Ohm
Static drain to source on-state resistance	3421	V_{GS} = 10 V dc, pulsed (see 4.5.1), condition A, I_D = rated I_{D1} (see 1.3)	$r_{DS(on)2}$			
2N7224, 2N7224U					0.081	Ohm
2N7225, 2N7225U					0.105	Ohm
2N7227, 2N7227U					0.415	Ohm
2N7228, 2N7228U					0.515	Ohm

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Forward voltage (source drain diode)	4011	Pulsed (see 4.5.1), condition A, $I_D = I_{D1}$ (see 1.3)	V_{SD}			
2N7224, 2N7224U					1.8	V
2N7725, 2N7225U					1.9	V
2N7227, 2N7227U					1.7	V
2N7228, 2N7228U					1.7	V
Subgroup 3						
High temperature operation:		$T_C = T_J = +125°C$				
Gate reverse current	3411	V_{GS} = +20 V dc and -20 V dc, bias condition C, V_{DS} = 0	I_{GSS2}		±200	nA dc
Drain current	3413	Bias condition C, V_{GS} = 0 V dc				
		V_{DS} = 80 percent rated	I_{DSS2}		0.25	mA dc
Static drain to source on-state resistance	3421	Condition A, V_{GS} = 10 V dc, pulsed (see 4.5.1) I_D = rated I_{D2} (see 1.3)	$r_{DS(on)3}$			
2N7224, 2N7224U					0.11	Ohm
2N7225, 2N7225U					0.17	Ohm
2N7227, 2N7227U					0.68	Ohm
2N7228, 2N7228U					0.90	Ohm
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$; I_D = .25 mA dc	$V_{GS(th)2}$	1.0		V dc
Low temperature operation:		$T_C = T_J = -55°C$				
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D = .25 mA dc	$V_{GS(th)3}$		5.0	V dc

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Switching time test	3472	$I_D = I_{D1}$ (see 1.3), V_{GS} = 10 V dc, Gate drive impedance = 2.35 ohms; $V_{DD} = 0.5\ V_{BR(DSS)}$				
Turn-on delay time			$t_{d(on)}$		35	ns
Rise time			t_r		190	ns
Turn-off delay time			$t_{d(off)}$		170	ns
Fall time			t_f		130	ns
Subgroup 5						
Safe operating area test	3474	See figure 8; V_{DS} = 80 percent of rated V_{DS} V_{DS} = 200 V maximum, t_p = 10 ms				
Electrical measurements		See table I, subgroup 2				
Subgroup 6						
Not applicable						
Subgroup 7						
Gate charge	3471	Condition B				
On-state gate charge			$Q_{g(on)}$			
2N7224, 2N7224U					125	nC
2N7225, 2N7225U					115	nC
2N7227, 2N7227U					110	nC
2N7228, 2N7228U					120	nC
Charge gate to source			Q_{gs}			
2N7224, 2N7224U					22	nC
2N7225, 2N7225U					22	nC
2N7227, 2N7227U					18	nC
2N7228, 2N7228U					19	nC

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 7 - Continued						
Charge gate to drain			Q_{gd}			
2N7224, 2N7224U					65	nC
2N7225, 2N7225U					60	nC
2N7227, 2N7227U					65	nC
2N7228, 2N7228U					70	nC
Reverse recovery time	3473	Condition A, $V_{DD} \leq 30$ V, $d_i/d_t \leq$ 100 A/µs, $I_D = I_{D1}$	t_{rr}			
2N7224, 2N7224U					500	ns
2N7225, 2N7225U					950	ns
2N7227, 2N7227U					1,200	ns
2N7228, 2N7228U					1,600	ns

1/ For sampling plan, see MIL-PRF-19500.
2/ For end-point measurements, this test is required for the following subgroups:
Group B, subgroups 2 and 3 (JANTXV).
Group B, subgroups 3 and 4 (JANS).
Group C, subgroup 2 and 6.
Group E, subgroup 1.

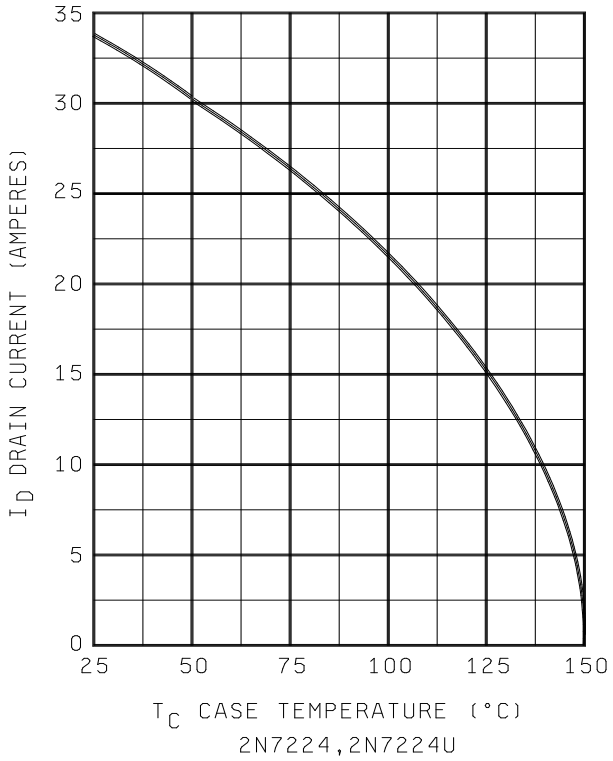
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TABLE II. Group E inspection (all quality levels) for qualification or re-qualification only.

Inspection 1/	MIL-STD-750		Sampling plan
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling	1051	500 cycles, test condition G	
Electrical measurements		See table I, subgroup 2	
Subgroup 2 2/			45 devices c = 0
Steady-state reverse bias	1042	Condition A, 1,000 hours	
Electrical measurements		See table I, subgroup 2	
Steady-state gate bias	1042	Condition B, 1,000 hours	
Electrical measurements		See table I, subgroup 2	
Subgroup 4			Sample size N/A
Thermal impedance curves		See MIL-PRF-19500	
Subgroup 5 3/			15 devices c = 0
Barometric pressure (reduced) 2N7227, 2N7227U 2N7228, 2N7228U	1001	Condition C, $V_{(ISO)} = V_{DS}$	
Subgroup 10			5 devices, c = 0
Repetitive avalanche energy	3469	Peak current IAR = ID; peak gate voltage VGS = 10 V; gate to source resistor, RGS 2.5 ≤ RGS ≤ 200 ohms, temperature = TJ = +150°C +0, -10°C Inductance = $\left[\frac{2E_{AR}}{(I_{D1})^2}\right]\left[\frac{V_{BR}-V_{DD}}{V_{BR}}\right]$ $mH\,\min$ Number of pulses to be applied = 3.6 X 108; supply voltage (VDD) = 50 V; time in avalanche = 2 μs minimum, 20 μs maximum; f = 500 Hz minimum	
Subgroup 11			22 devices, c = 0
Commutating diode for safe operating area test procedure for measuring dv/dt during reverse recovery of power MOSFET transistors or insulated gate bipolar transistors	3476	Test conditions shall be derived by the manufacturer	

1/ JANHC and JANKC device are qualified with MIL-PRF-19500.
2/ A separate sample for each test may be pulled.
3/ Not required for 2N7224, 2N7224U, 2N7225, and 2N7225U.

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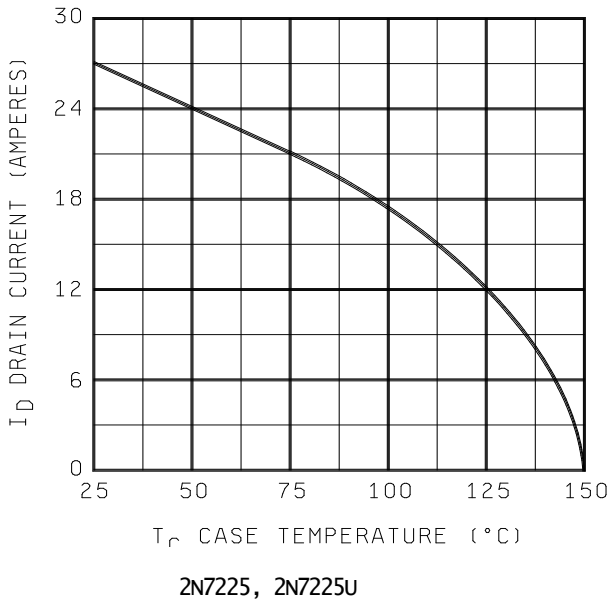
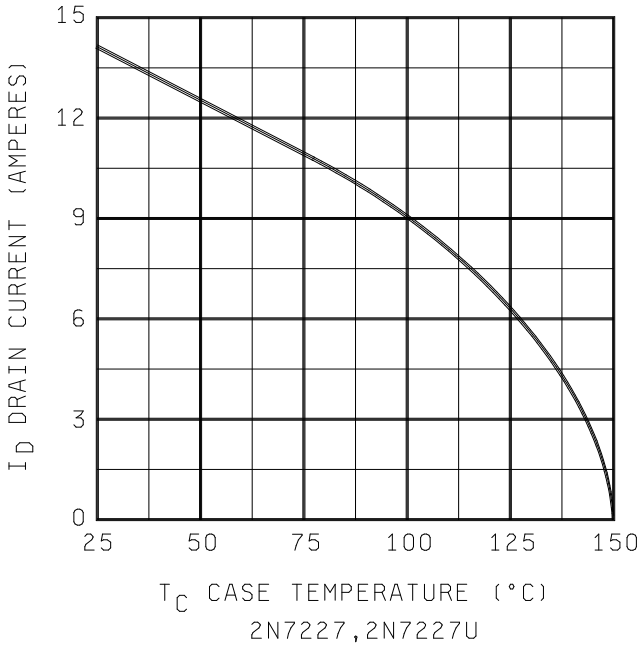


FIGURE 6. Maximum drain current versus case temperature.

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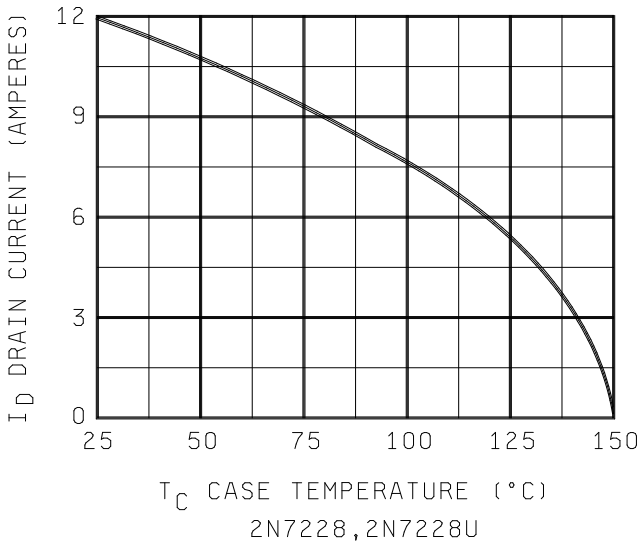


FIGURE 6. Maximum drain current versus case temperature - Continued.

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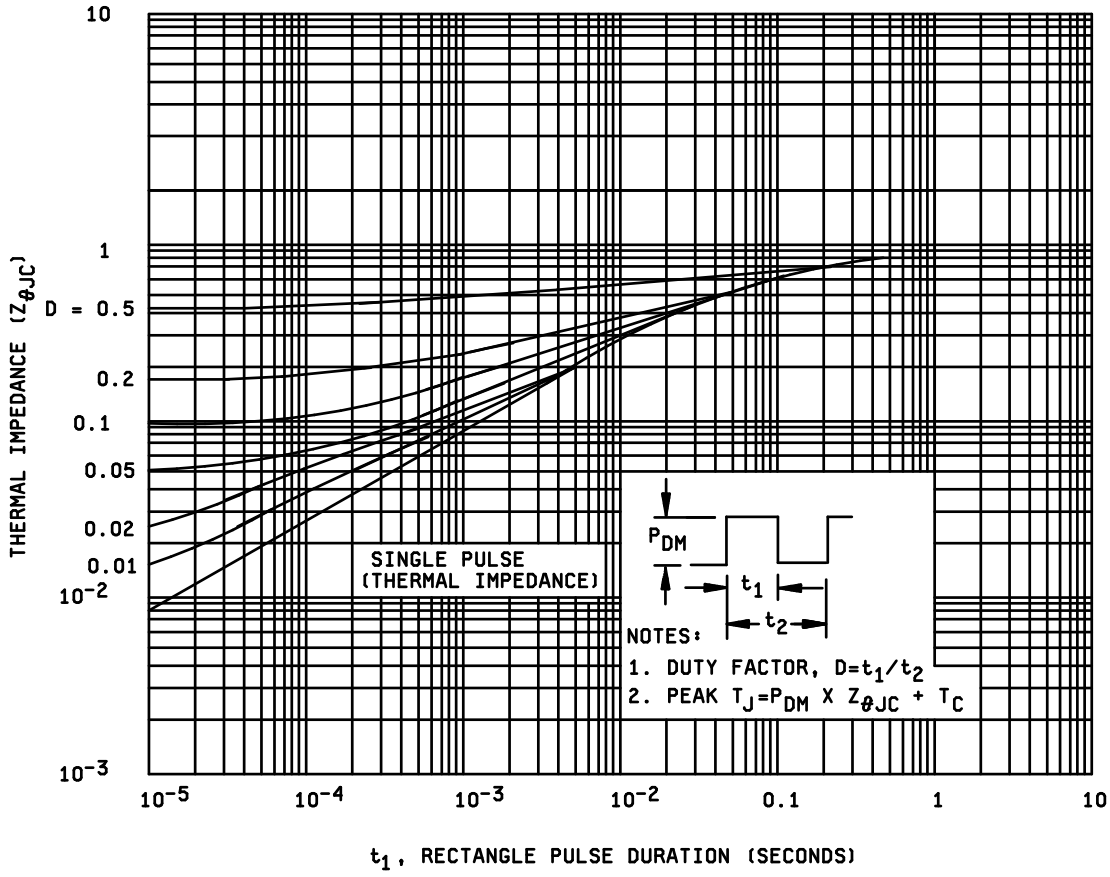
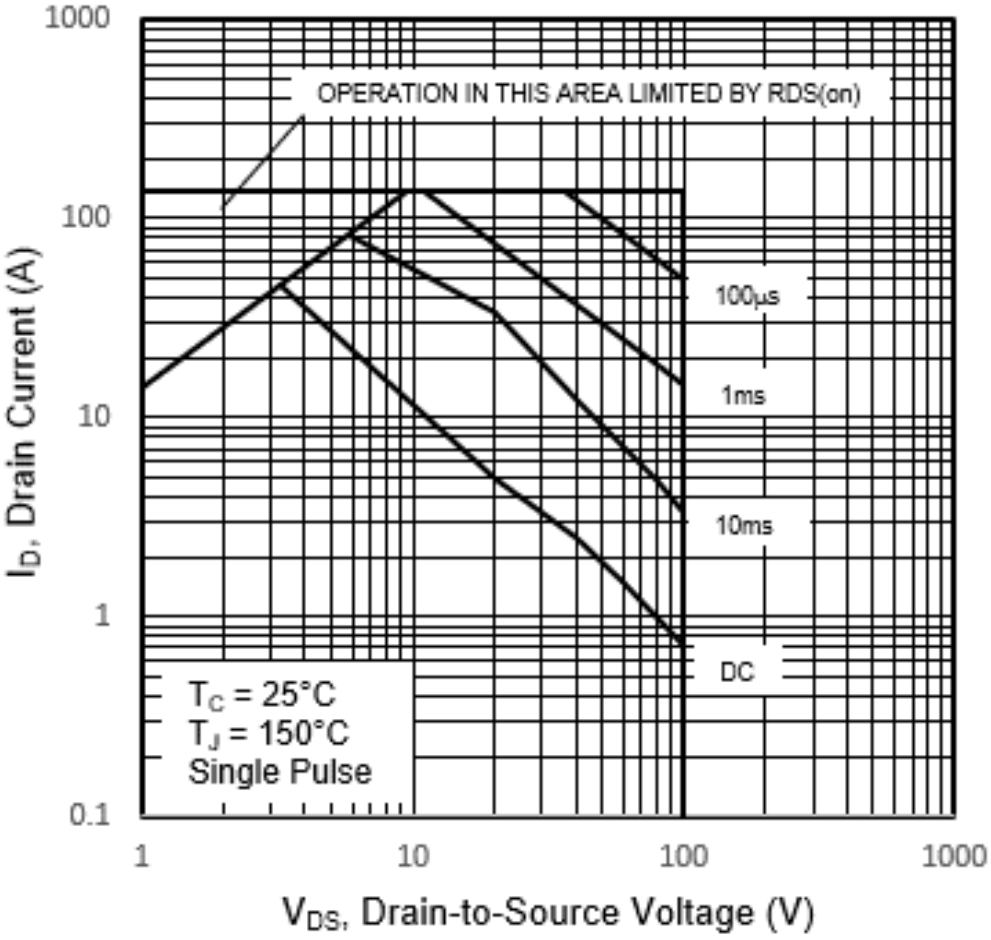


FIGURE 7. Thermal impedance curves.

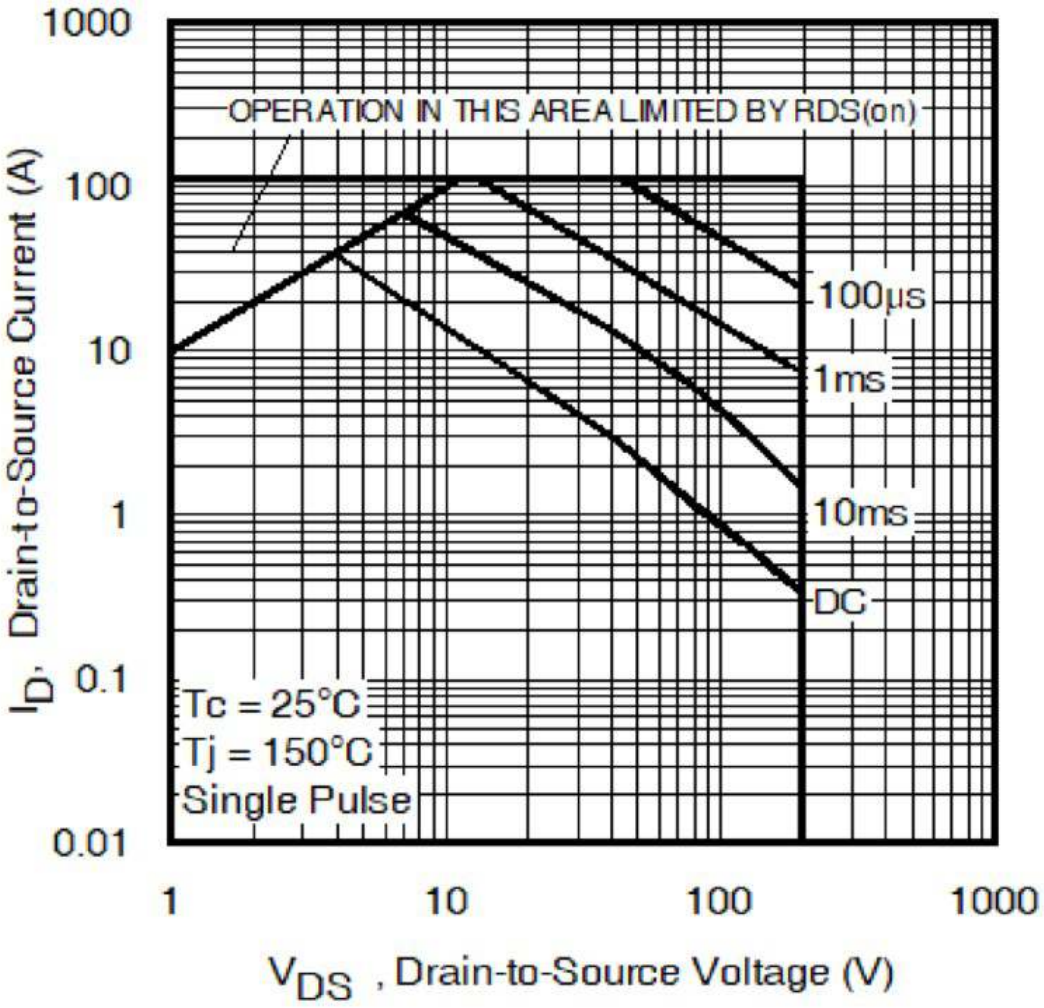
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2N7224, 2N7224U

* FIGURE 8. Safe operating area graph.

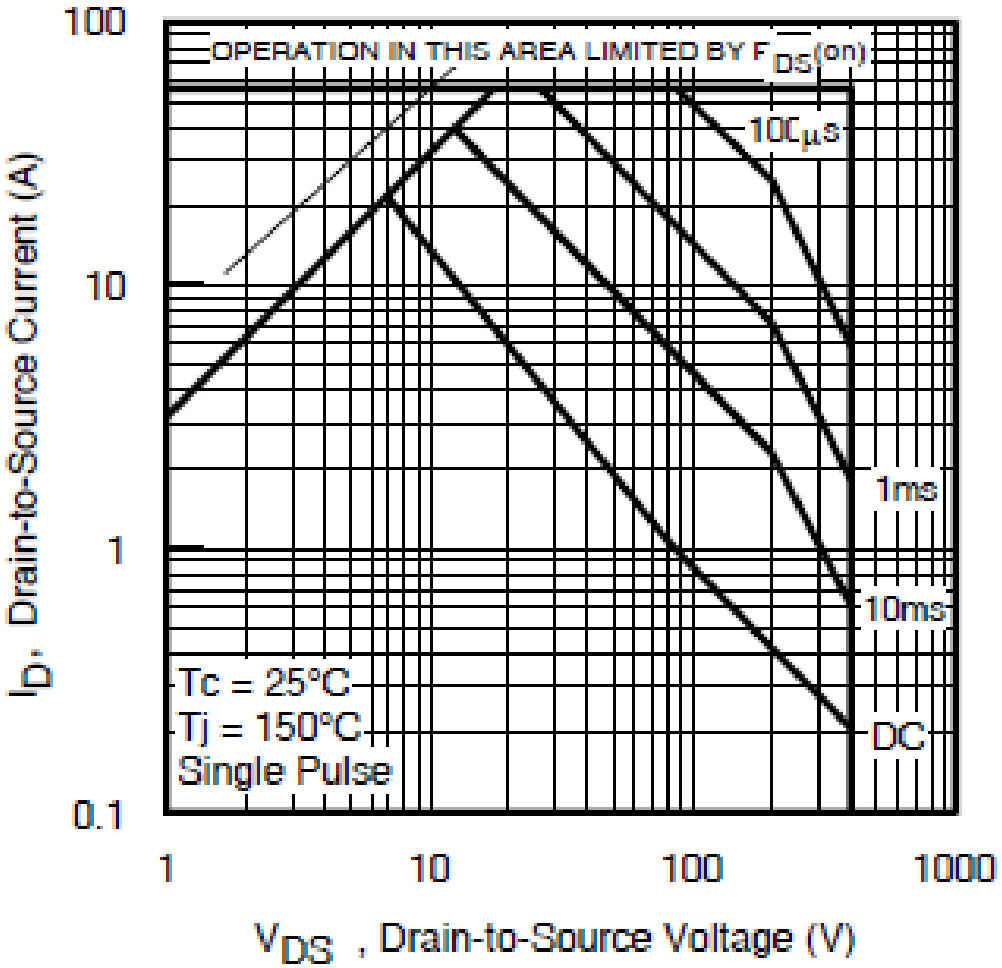
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2N7225, 2N7225U

FIGURE 8. Safe operating area graph - Continued.

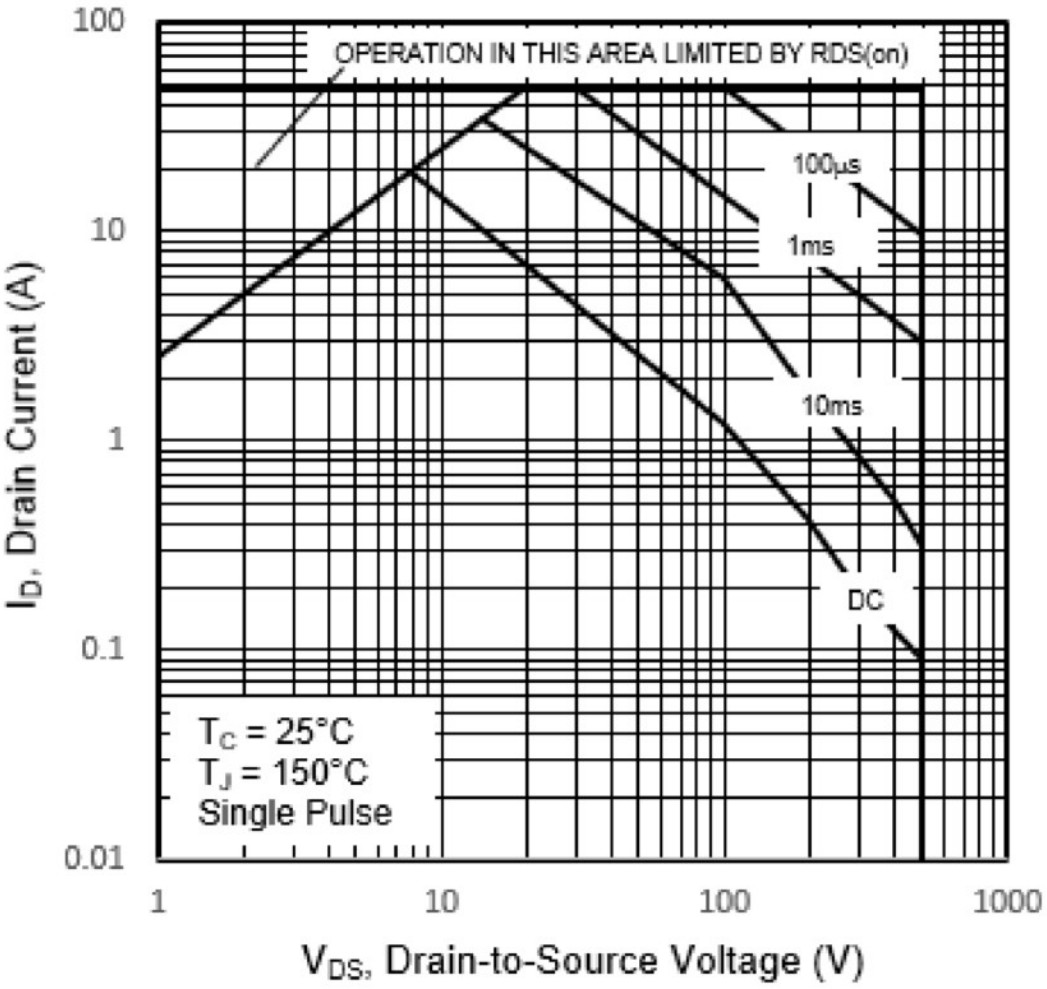
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2N7227, 2N7227U

* FIGURE 8. Safe operating area graph - Continued.

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2N7228, 2N7228U

FIGURE 8. Safe operating area graph - Continued.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. The complete PIN, see 1.5 and 6.5.

e. For die acquisition, the JANHC or JANKC letter version shall be specified (see figures 3, 4, and 5).

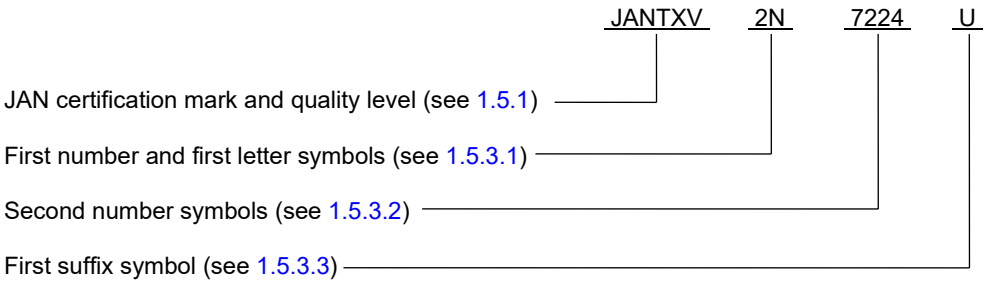
6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.4 Supersession and substitution of DESC drawing. This specification supersedes DESC drawing 89026, dated 19 December 1989.

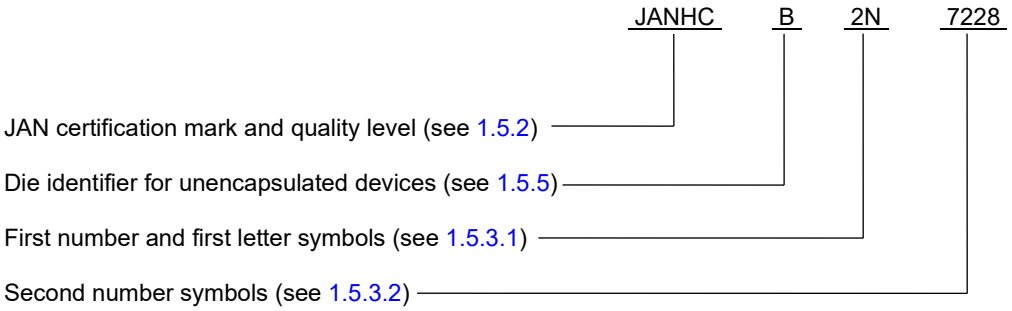
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6.5 PIN construction example.

6.5.1 Encapsulated devices The PINs for encapsulated devices are constructed using the following form.



6.5.2 Unencapsulated devices. The PINs for un-encapsulated devices are constructed using the following form.



6.6 List of PINs.

6.6.1 List of PINs for encapsulated devices. The following is a list of possible PINs for encapsulated devices available on this specification sheet.

PINs for devices of the base quality level	PINs for devices of the "TX" quality level	PINs for devices of the "TXV" quality level	PINs for devices of the "S" quality level
JAN2N7224	JANTX2N7224	JANTXV2N7224	JANS2N7224
JAN2N7224U	JANTX2N7224U	JANTXV2N7224U	JANS2N7224U
JAN2N7225	JANTX2N7225	JANTXV2N7225	JANS2N7225
JAN2N7225U	JANTX2N7225U	JANTXV2N7225U	JANS2N7225U
JAN2N7227	JANTX2N7227	JANTXV2N7227	JANS2N7227
JAN2N7227U	JANTX2N7227U	JANTXV2N7227U	JANS2N7227U
JAN2N7228	JANTX2N7228	JANTXV2N7228	JANS2N7228
JAN2N7228U	JANTX2N7228U	JANTXV2N7228U	JANS2N7228U

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6.6.2 List of PINs for unencapsulated devices. The following is a list of possible PINs for unencapsulated devices available on this specification sheet. The qualified die suppliers will be identified on the QML (example JANHCA7224).

JANC ordering information		
PIN	Manufacturers	
	69210	43611
2N7224	JANHCA2N7224 JANKCA2N7224	JANHCB2N7224 JANKCB2N7224
2N7225	JANHCA2N7225 JANKCA2N7225	JANHCB2N7225 JANKCB2N7225
2N7227	JANHCA2N7227 JANKCA2N7227	JANHCB2N7227 JANKCB2N7227
2N2778	JANHCA2N7228 JANKCA2N7228	JANHCB2N7228 JANKCB2N7228

6.7 Request for new types and configurations. Requests for new device types or configurations for inclusions in this specification sheet should be submitted to: DLA Land and Maritime, ATTN: VAC, Post Office Box 3990, Columbus, OH 43218-3990 or by electronic mail at Semiconductor@dla.mil or by facsimile (614) 692-6939 or DSN 850-6939.

6.8 Amendment notations. The margins of this specification are marked with asterisks to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army - CR
Navy - SH
Air Force - 85
NASA - NA
DLA - CC

Review activities:
Army - AR, MI
Navy - AS
Air Force – 19, 70, 170

Preparing activity:
DLA - CC

(Project 5961-2024-008)

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